

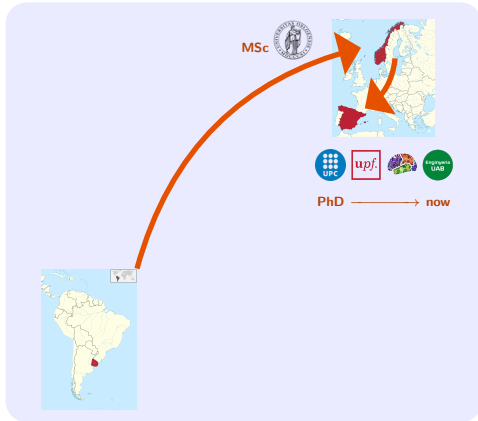
EEG Datasets

BioRTC 2026 - Yobe State University

Silvana Silva Pereira

June 2026

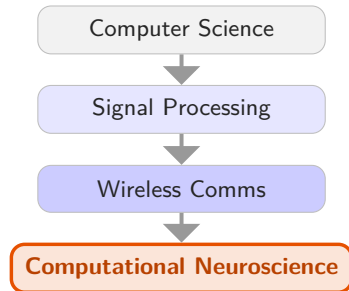
The journey



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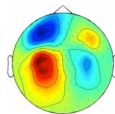
Following questions



What do I work on?

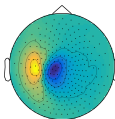
EEG

Oscillations, connectivity & time-frequency analysis of language and cognition



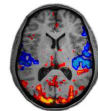
MEG

Source-level spatiotemporal dynamics; high-density sensor arrays



fMRI

Language network mapping; BOLD mediation modelling



My main focus: Non-Invasive Brain Imaging & Language

What Data Will We Work With?

TUH EEG Corpus (Temple University)

Three public sub-corpora from a single clinical EEG bank:

- **TUAR** v3.0.1 *Artifact annotations*
- **TUSZ** v2.0.6 *Seizure labels*

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Project track 1	TUSZ	DL classification
Project track 2	TUSZ	Source reconstruction
Both projects	YSUTH EDF	Transfer / inference

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Local Clinical Recording

SUBJECT1.edf

Yobe State University Teaching Hospital (YSUTH) - real patient, no ground-truth labels

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Important

All three share the same EDF format and referential storage - pipelines transfer across them.

For this course

The TUH EEG data is shared under a **Data Use Agreement (DUA)** signed by me covering all participants.

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Download it yourself

Register at the NEDC portal - free, academic use:

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Once approved you can access TUAR, TUSZ, TUEP and the full corpus via rsync.

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Course materials

Notebooks and slides:

https://github.com/comptelab/BIORTC_Nigeria

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Demo file we will use

aaaaabbn_s010_t000.edf

305 s, 19 channels

Events in first 120 s:

- 3-36 s: elec on C3-CZ
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Why it matters

Real artifact labels let us ask: *can ICA remove all of these?* (Spoiler: not always.)

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Key challenge

Seizures are **rare events** ($< 5\%$ of recording time). Class imbalance is not just a stats problem - it changes everything about how you build and evaluate the model.

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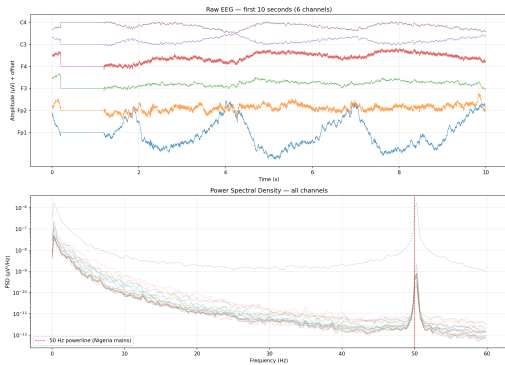
The real challenge

Your model is trained on labeled TUH data. Can it generalize to a different hospital, equipment, and sampling rate?

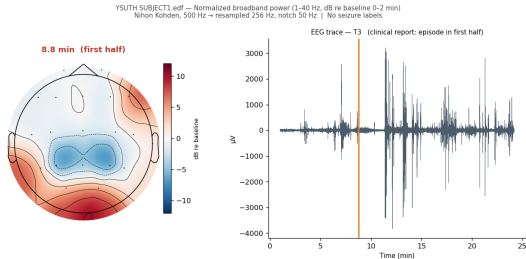
This is the **domain adaptation** problem in clinical AI.

Note: The recording is anonymized.

The Local Data - What It Looks Like



Raw trace + PSD



Normalized power topomap, full 24-min recording at
16 $\times$ speed

[Click to play \(yobe_topomap.mp4\)](#)